D Karthik Reddy

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SUMMARY

Final-year B.Tech graduate (May 2026, CGPA: **8.97**) and aspiring **Software Engineer** with strong foundations in **data structures, algorithms, object-oriented design**, and full-stack development. Proficient in **Python, C, C++, JavaScript, and SQL** with 183 LeetCode problems solved. Built and deployed multiple production-grade applications with REST APIs, automated testing, and CI/CD. IEEE-published author, patent co-inventor, and Smart India Hackathon 2023 national winner. Passionate about building reliable, scalable software that solves real-world problems.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology — Hyderabad, India
B.Tech in Electronics & Communication Engineering | CGPA: 8.97 | Minor in AI & ML | CGPA: 8.5 — *2022 – May 2026*

PROJECTS

AI GitHub PR Reviewer (Live: ai-pr-reviewer-eta.vercel.app) — **2026**
Python | FastAPI | React | REST APIs | Git | CI/CD | Unit Testing

- Designed and built a **full-stack application** with a 5-stage backend processing pipeline – architected with object-oriented design, REST API endpoints, rate limiting, and structured error handling deployed to production.
- Implemented **automated testing, debugging, and code reviews**; applied 16 production hardening fixes with descriptive Git commit history and CI/CD auto-deployment on every push.
- Analyzed a real **Facebook React codebase** (31k stars) in **8.3 seconds**, identifying 6 accurate issues with file-level precision – demonstrating ability to build performant, scalable software.

AI Email Agent (Live: ai-email-agent-alpha.vercel.app) — **2026**
Python | FastAPI | React | REST APIs | SQL | Git | Unit Testing

- Architected a **scalable backend system** with object-oriented design – 7 REST endpoints, persistent SQL state management, rate limiting, input validation, and full audit logging.
- Built **4 independently testable processing modules** communicating through shared state, enabling fault isolation; applied unit testing and structured debugging throughout development.
- Delivered **end-to-end processing under 10 seconds** with a responsive React frontend and human-in-the-loop validation; deployed full-stack with CI/CD automation.

UAV Magnetic Anomaly Detection – Smart India Hackathon 2023 — **2023**
Python | C | C++ | Embedded Systems | Real-Time Systems | Algorithm Design

- Led a cross-functional team to design and deliver a **real-time embedded software system** for the Ministry of Defence – managed requirements, system design, and delivery under tight national hackathon constraints.
- Wrote **low-level C and C++ algorithms** for real-time sensor data acquisition; performed complexity analysis to optimize processing pipelines for real-time performance constraints.
- Achieved **40% improvement in detection accuracy** and **25% increase in drone flight time**; awarded **1st place at Smart India Hackathon 2023**, Ministry of Defence track – 1M+ participants nationwide.

Ensemble Vision Transformers – GI Disease Classification — **2025**
Python | PyTorch | C++ | Algorithm Design | Scikit-learn | Git

- Designed a **multi-model ensemble algorithm** combining 3 vision transformers via a logistic regression meta-classifier; performed complexity and tradeoff analysis across accuracy, latency, and cost on 8,000 images.
- Achieved **97.75% accuracy**, outperforming single-model baselines by 4.2%; research accepted at **IEEE ICIIP 2025** – peer-reviewed international conference.

TECHNICAL SKILLS

Languages:	Python, C, C++, JavaScript, SQL, HTML, CSS
Computer Science:	Data Structures & Algorithms (**183 LeetCode**), Object-Oriented Design & Programming, Complexity Analysis, Algorithm Design, System Design (learning)
Web & Development:	React, FastAPI, Node.js, REST APIs, PostgreSQL, OAuth 2.0
Tools & Practices:	Git, GitHub, CI/CD (GitHub Actions), Unit Testing, Code Review, SDLC, Agile
Core CS:	Operating Systems, DBMS, Computer Networks, Distributed Systems fundamentals

AI / ML: PyTorch, TensorFlow, LangGraph, LangChain, Scikit-learn, Computer Vision

ACHIEVEMENTS

IEEE Publication (ICIIP 2025): Lead author – Vision Transformers achieving **97.75% accuracy** on GI disease classification; peer-reviewed international conference

Smart India Hackathon 2023 – National Winner: Ministry of Defence track; 40% detection accuracy improvement, 25% flight time increase; 1M+ participants nationwide

Patent Filed: Co-inventor of AI-based acoustic real-time rail defect detection system for predictive maintenance